📘 Lecture 1 – Software Construction

30 MCQs (Dr. Ehab Essa Style)

1.

Software Construction mainly focuses on:
a) Requirements
b) Design
c) Coding
d) Planning

Answer: c) Coding

2.

Which activity belongs directly to Software Construction?
a) Debugging
b) Feasibility study
c) Marketing
d) Budgeting

Answer: a) Debugging

3.

The primary output of Software Construction is:
a) UML
b) Source code
c) Specification
d) Schedule

Answer: b) Source code

4.

Correct behavior at the present time refers to:
a) Robustness
b) Correctness
c) Maintainability
d) Portability

Answer: b) Correctness

5.

Correct behavior in the future mainly refers to:
a) Correctness
b) Defensiveness
c) Performance
d) Efficiency

Answer: b) Defensiveness

6.

Software that is easy to understand mainly supports:
a) Compiler
b) Hardware
c) Maintenance
d) Deployment

Answer: c) Maintenance

7.

“Ready for change” software is best described as:
a) Stable
b) Extensible
c) Secure
d) Portable

Answer: b) Extensible

8.

Reducing complexity mainly improves:
a) Speed
b) Readability
c) Storage
d) Compilation

Answer: b) Readability

9.

Hiding internal details from users is called:
a) Encapsulation
b) Abstraction
c) Inheritance
d) Polymorphism

Answer: b) Abstraction

10.

Breaking a system into independent parts is known as:
a) Refactoring
b) Modularity
c) Debugging
d) Integration

Answer: b) Modularity

11.

Which concept mainly supports easier change?
a) Optimization
b) Anticipation
c) Compilation
d) Execution

Answer: b) Anticipation

12.

Constructing for verification mainly helps:
a) Performance
b) Fault detection
c) UI design
d) Portability

Answer: b) Fault detection

13.

Coding standards mainly improve:
a) Speed
b) Readability
c) Memory
d) Accuracy

Answer: b) Readability

14.

Which is an example of an external standard?
a) Team rules
b) Project notes
c) IEEE
d) Comments

Answer: c) IEEE

15.

Reuse means using:
a) New code
b) Existing assets

c) Temporary code
d) Generated code

Answer: b) Existing assets

16.

Construction *with* reuse means:
a) Creating libraries
b) Writing frameworks
c) Using assets
d) Copy-paste

Answer: c) Using assets

17.

Construction *for* reuse means:
a) Using APIs
b) Creating reusable code
c) Testing reuse
d) Removing code

Answer: b) Creating reusable code

18.

Which model is strictly linear?
a) Agile
b) Spiral
c) Waterfall
d) Incremental

Answer: c) Waterfall

19.

Agile development is mainly:
a) Sequential
b) Linear
c) Iterative
d) Predictive

Answer: c) Iterative

20.

Construction planning mainly defines:
a) Syntax
b) Algorithms
c) Tasks & schedule
d) UI

Answer: c) Tasks & schedule

21.

Which metric measures number of independent paths?
a) LOC
b) Cyclomatic complexity
c) Defect density
d) Coupling

Answer: b) Cyclomatic complexity

22.

Defect density is a measure of:
a) Time
b) Cost
c) Quality
d) Size

Answer: c) Quality

23.

Which is a construction language?
a) UML
b) Java
c) BPMN
d) ER

Answer: b) Java

24.

HTML is classified as:
a) Programming
b) Construction
c) Markup
d) Scripting

Answer: c) Markup

25.

Unit testing is mainly performed during:
a) Design
b) Construction
c) Deployment
d) Maintenance

Answer: b) Construction

26.

Incremental integration mainly helps in:
a) Late testing
b) Easier debugging
c) No testing
d) Higher risk

Answer: b) Easier debugging

27.

Which activity ensures software works as intended?
a) Debugging
b) Testing
c) Planning
d) Scheduling

Answer: b) Testing

28.

Software Construction is closest to which phase?
a) Requirements
b) Design
c) Implementation
d) Maintenance

Answer: c) Implementation

29.

Which improves consistency among developers?
a) Refactoring
b) Standards
c) Optimization
d) Compilation

Answer: b) Standards

30.

The main goal of Software Construction is delivering:
a) Diagrams
b) Documents
c) Working software
d) Schedules

Answer: c) Working software

📝 Essay Question 1

Explain the concept of Software Construction.
Why is it considered a critical phase in the software development process?

💡 ***In your answer, you are expected to discuss:***

- **Definition of Software Construction**
- **Main activities (coding, debugging, testing, integration)**
- **Why construction quality directly affects correctness, maintainability, and future change**

📝 Essay Question 2

"Good software must be correct, easy to understand, and ready for change."
Explain this statement in the context of Software Construction.

💡 ***In your answer, you are expected to discuss:***

- **Correctness (present behavior)**
- **Defensiveness / readiness for future change**
- **Readability and maintainability**
- **How construction practices support these goals**

📝 Essay Question 3

Discuss the importance of standards and reuse in Software Construction.

💡 ***In your answer, you are expected to discuss:***

- **Coding standards and their benefits**
- **Construction with reuse vs construction for reuse**
- **How reuse and standards reduce cost, errors, and development time**